

Bowen Feng

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Final-year MInf student at the University of Edinburgh working on 3D reconstruction and computational imaging: feed-forward geometry at mirrors and glass, sparse-view generative reconstruction, and physics-based imaging.

EDUCATION

University of Edinburgh

2022 - 2027 (expected)

Master of Informatics (MInf), five-year integrated master's program, School of Informatics, Edinburgh, UK. Overall average 70/100.

Relevant coursework: Computer Graphics: Rendering, Computer Graphics: Geometry and Simulation, Computer Vision, Machine Learning, Deep Learning.

University of California, Davis

2024 - 2025

Exchange student, Davis, CA, United States (GPA 3.82/4.0)

RESEARCH INTERESTS

Feed-forward and generative 3D reconstruction; reflective and transparent surfaces and the identifiability of geometry; physics-based imaging (exposure, blur, microscopy); reliability of 3D and video perception.

SELECTED RESEARCH PROJECTS

Reliable 3D Reconstruction at Mirrors and Glass

First-author research | Computer vision

- A planar mirror's virtual scene is rigid across views, has positive depth and gets high confidence, so feed-forward 3D models reconstruct it as geometry. Collected CounterRay: 22 indoor scenes with 38 mirror or glass planes, each captured exposed, covered and from the physical side, which gives separate ground truth for the interface and for the reflected content.
- In an audit of 200 reflective regions from ScanNet++ and ARKitScenes, 24.1% of the samples accepted by released validity masks follow reflected content rather than the surface. One mirror-dominated frame raises VGGT pose error on otherwise clean frames by up to 2.42x. A light-path pointmap model that predicts path type, plane support and physical endpoint cuts median interface/content error from 10.9/31.8 cm to 4.8/15.9 cm.

OneScene: Observation-Adaptive Scene Spaces for Sparse-View Reconstruction

First-author research | 3D vision

- A generative reconstruction model whose continuous function space adapts to the observations: an image-conditioned mean and residual dictionary represent geometry and appearance together, and a conditional flow generates their coefficients before any mesh or image is requested.

Underwater Instance Segmentation with Teacher-Aligned Reliability (TIDE-IS)

Sole-author research | Marine perception

- A 28,801-parameter reliability adapter on a lightweight detection transformer; five-seed mask AP on USIS10K rises from 66.80 to 70.13 and matched-boundary F-score from 72.6 to 76.0.

PUBLICATIONS

TIDE-IS: Teacher-Aligned Instance Reliability and Bounded Contour Refinement for Underwater Instance Segmentation.

Bowen Feng (sole author).

International Conference on Intelligent Sensing, Communications and Marine Applications (ISCMA 2026), accepted.

Micro-Expression Recognition System Based on Dual-Stream Graph Structure and Action Unit Auxiliary Learning.

Bowen Feng (sole author).

International Conference on Intelligent Sensing, Communications and Marine Applications (ISCMA 2026), accepted.

Probabilistic Dual-Adaptive Spatio-Temporal Graph Convolutional Networks for Forecasting Energy Consumption Dynamics of Electric Vehicle Charging Stations.

Djamel Eddine Mekkaoui, Mohamed Amine Midoun, Abdelkarim Smaili, **Bowen Feng**, Mohamed Zakariya Talhaoui, and Yanming Shen.

Computers & Electrical Engineering, 122, 109976 (2025). doi.org/10.1016/j.compeleceng.2024.109976

SPR: A Similar Projection Revisor for Complex Logical Reasoning over Knowledge Graphs.

Tingting Wang, Yuxuan Tang, Ruolin Li, Duo Yu, **Bowen Feng**, Feng Ding, Shuo Yu, and Yanming Shen.

Web Information Systems and Applications (WISA 2024), LNCS 14883, pp. 78-90. doi.org/10.1007/978-981-97-7707-5_7

UNDER REVIEW

When Should a Text-Prompted Segmenter Abstain? Area-Limited Foreground-Context Evidence for One-Prompt Presence Decisions.

Bowen Feng. Under review (journal), 2026.

MANUSCRIPTS IN PREPARATION

Beyond Straight Rays: Counterfactual Auditing of Feed-Forward 3D Geometry at Mirrors and Glass *First author.*

Manuscript in preparation, 2026.

OneScene: Observation-Adaptive Scene Spaces for Joint Geometry and Appearance *First author. Manuscript in*

preparation, 2026.

RESEARCH EXPERIENCE

Research with Prof. Tian Xie, Zhejiang University 2025 - 2026

Auditable web workflows (TTAT): system design, WebArena evaluation and a 12-user study.

Research assistant, Prof. Yanming Shen's group, Dalian University of Technology 2022 - 2024

Knowledge-graph reasoning (WISA 2024) and spatio-temporal forecasting (Computers & Electrical Engineering 2025): data processing, temporal modeling components and baseline comparisons.

Bioinformatics and Statistical Analysis Intern, CAICT, Beijing 2025

Prepared data and covariates in R and Python; model selection, multiple-comparison control, effect sizes and confidence intervals; sensitivity analyses and reproducible reports.

SERVICE

Student Representative (Year 4), School of Informatics, University of Edinburgh 2025 - 2026

Programme Representative, Computer Science, School of Informatics, University of Edinburgh 2023 - 2024

Year 2 Student Representative, School of Informatics, University of Edinburgh 2023 - 2024

SKILLS

Programming: Python, R, C++, JavaScript, LaTeX.

Frameworks and tools: PyTorch, OpenCV, NumPy/SciPy, scikit-learn, pandas, Playwright and Selenium for browser automation, Git; training on multi-GPU clusters.

Methods: Controlled and counterfactual evaluation, paired-capture datasets, calibration and conformal prediction, ablation studies, user studies with statistical inference, failure analysis.